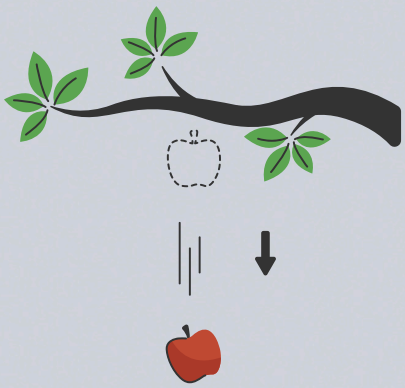


Group 8

GREEN INTERVENTION FOR SUSTAINABLE CAPSTONE THESIS SUBMISSION: REPLACING PRINTED BLACK BOOKS WITH A DIGITAL REPOSITORY



CASE STUDY 2
AUGUST 2026

ACKNOWLEDGEMENTS

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THE PROCESS



Each semester, around 200 students, or roughly 50 Capstone C groups, complete their reports/black book/thesis and must print at least two copies, while some groups print three copies. With reports ranging from about 50 to 100 pages or more, this means approximately 5,000 to 15,000 pages may be printed in a single semester. At the stated local colour-printing rate of BDT 10 per page and BDT 60 for binding, students collectively spend around BDT 56,000 to BDT 159,000. To reduce this expense, many groups travel to Nilkhet, where printing costs about BDT 2.50 per page and binding about BDT 10–20, reducing the printing and binding cost to roughly BDT 13,500-40,500, excluding transport. However, students must travel by bus, CNG, car or rickshaw, adding cost, time, traffic exposure and carbon emissions.

GREEN GAP ANALYSIS

ESTIMATED PAPER-PRODUCTION FOOTPRINT

Using standard 80-gsm A4 paper, one sheet weighs approximately 4.99 grams. Therefore, 5,000 sheets weigh about 24.95 kilograms, while 15,000 sheets weigh about 74.84 kilograms. Using the 2026 UK Government primary-paper emission factor of 1.34359114 kilograms of CO_2e per kilogram of paper, 5,000 sheets produce approximately 33.52 kilograms of CO_2e, while 15,000 sheets produce approximately 100.57 kilograms of CO_2e. Overall, printing 5,000 to 15,000 pages requires approximately 25 to 75 kilograms of paper and creates an estimated paper-production footprint of around 34 to 101 kilograms of CO_2e per semester.



ESTIMATED PRINTING-ELECTRICITY EMISSIONS

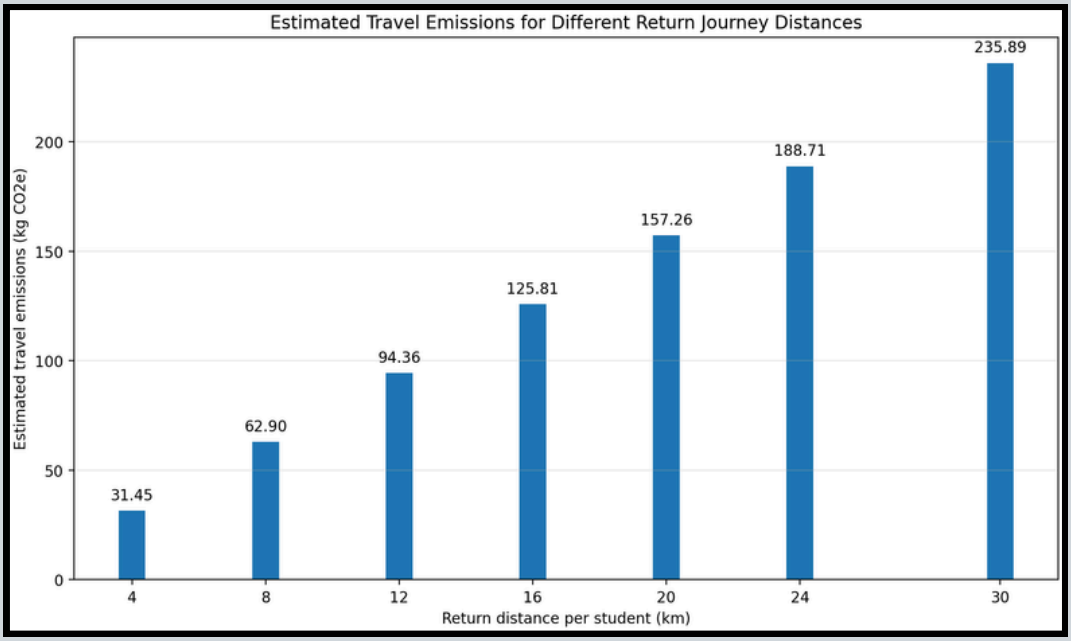
The HP Color LaserJet Enterprise 5800zf consumes approximately 655 watts, or 0.655 kilowatts, while printing and can print up to 43 A4 pages per minute. Based on these specifications, printing 5,000 pages would take approximately 1.94 hours and consume about 1.27 kWh of electricity, while printing 15,000 pages would take approximately 5.81 hours and consume about 3.81 kWh. Applying a grid-emission factor of 0.648 kg CO_2e per kWh, the associated

printing-electricity emissions would range from approximately 0.82 to 2.47 kg CO_2e per semester. These estimates include active printer electricity only and exclude warm-up, idle operation, shop lighting, computers, cooling, binding equipment, and reprinting.

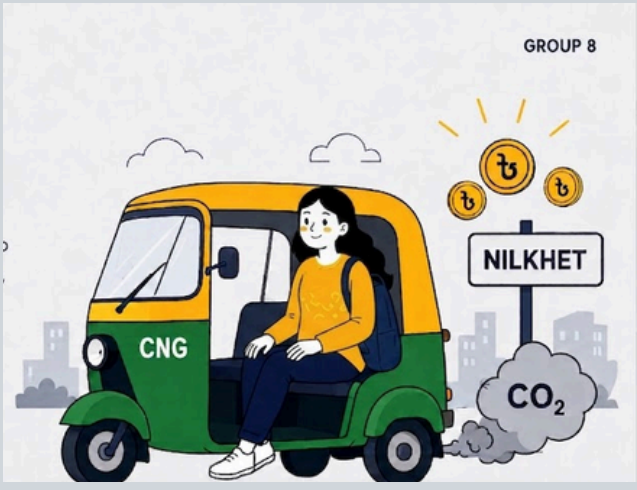
ESTIMATED WATER FOOTPRINT

The Water Footprint Network estimates that producing printing and writing paper requires approximately 2 to 13 litres of supply-chain water per A4 sheet, depending on factors such as the paper source, recycled content, paper type, and manufacturing process. This water footprint includes water used in forestry, pulp production, and paper manufacturing, rather than only water used at the printing shop. Based on this estimate, producing 5,000 sheets of paper could require approximately 10,000 to 65,000 litres of water, while producing 15,000 sheets could require approximately 30,000 to 195,000 litres. Therefore, the estimated overall supply-chain water footprint of printing 5,000 to 15,000 pages is approximately 10,000 to 195,000 litres per semester.

ESTIMATED TRANSPORT-RELATED EMISSION

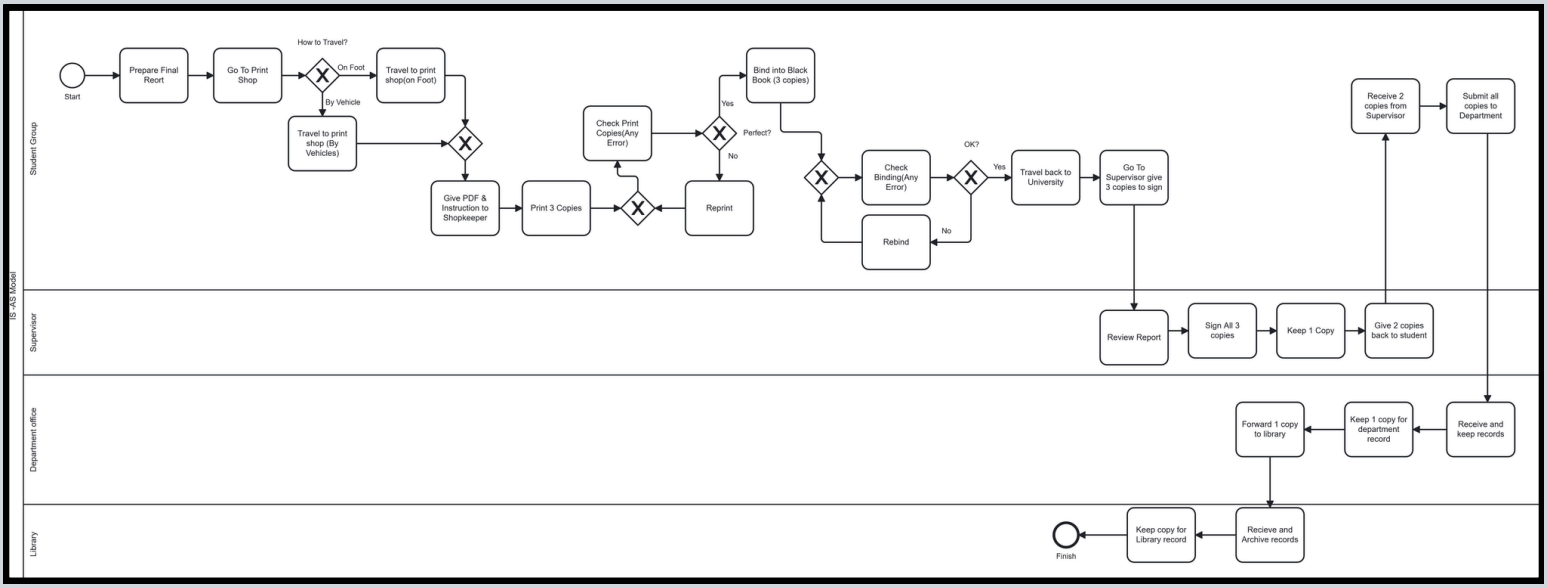


Using the UK Government's 2026 conversion factors, a non-London local bus produces approximately 0.12552 kg CO_2e per passenger-kilometre. After adding the fuel-supply or well-to-tank factor of 0.03174 kg CO_2e per passenger-kilometre, the combined planning factor becomes 0.15726 kg CO_2e per passenger-kilometre. Assuming one student from each of 50 Capstone groups makes one return journey to a printing location, a 4-km return trip would create approximately 31.45 kg CO_2e, while an 8-km journey would produce 62.90 kg CO_2e. The estimated emissions would rise to 94.36 kg CO_2e for 12 km, 125.81 kg CO_2e for 16 km, 157.26 kg CO_2e for 20 km, 188.71 kg CO_2e for 24 km, and 235.89 kg CO_2e for 30 km.



For example, if one student from each group travels a 16-km return distance by bus, the combined distance would be 800 passenger-kilometres and the estimated travel footprint would be approximately 125.81 kg CO_2e, or about 126 kg CO_2e per semester. This figure uses an international bus-emission factor as a planning proxy rather than a measured Dhaka-specific value. Actual emissions may vary depending on the vehicle type, fuel, traffic congestion, journey distance, route conditions, and passenger occupancy. These results show that the current physical submission process creates not only a financial burden for students but also avoidable paper waste, water consumption, carbon emissions, binding-material waste, transport-related pollution.

AS-IS MODEL



The AS-IS model represents the current physical Capstone C thesis submission process. After the supervisor approves the final report, the student group prepares the PDF and decides whether to print at a nearby local shop or travel to Nilkhet for lower printing costs. The group then submits the file to the printing shop, where two or three copies are printed and bound as black books.

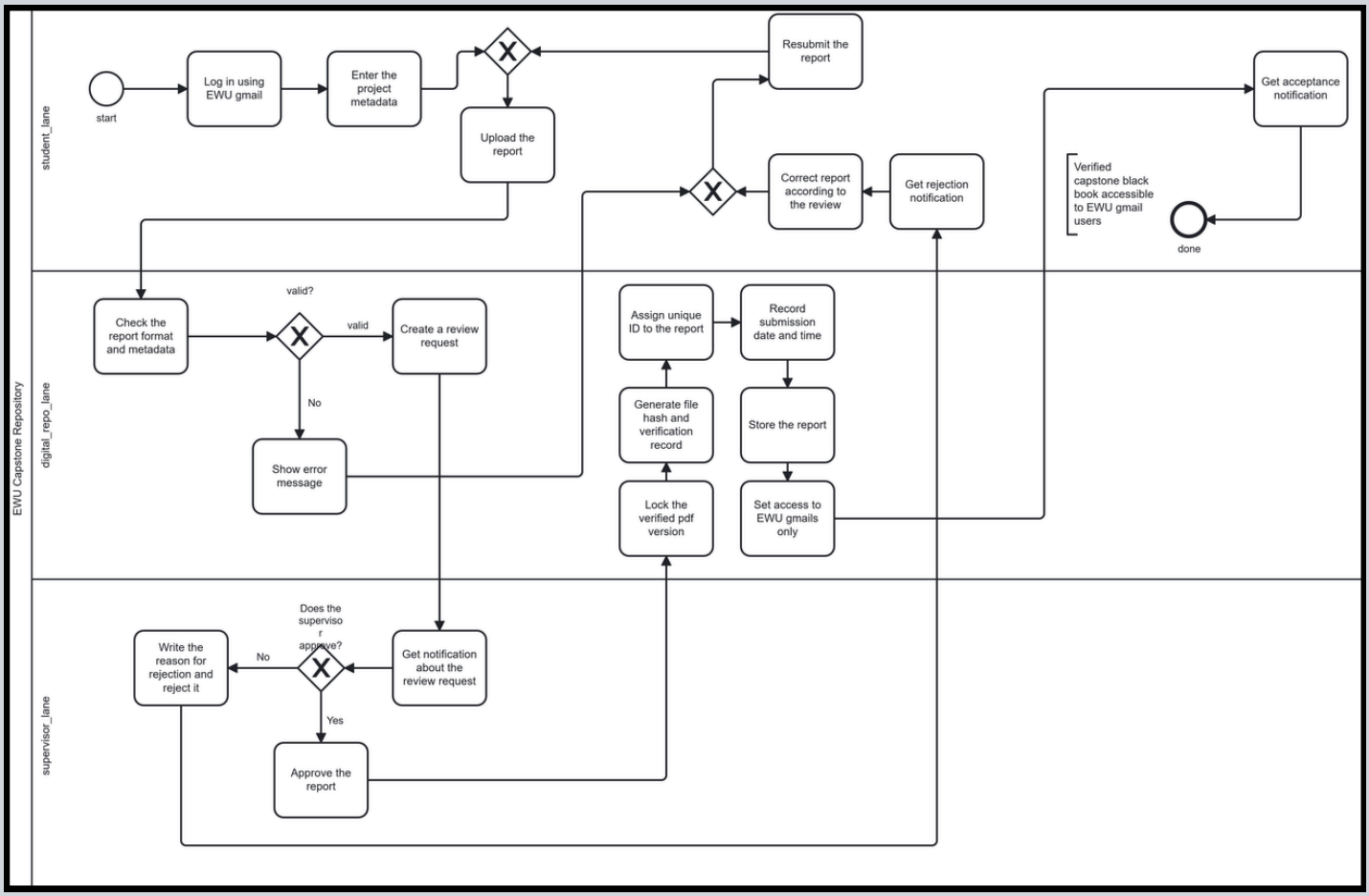
After printing, the students inspect the copies for missing pages, incorrect page order, poor image quality, colour problems, or binding defects. When an error is found, the affected pages or the entire copy must be reprinted and checked again, increasing waiting time, electricity use, cost, and paper waste. Once all copies are correct, the students collect the black books, return to the university, obtain any required signatures, and physically submit the copies to the supervisor, department, and library.

The model highlights several weaknesses in the current process, including repeated printing, student travel, long waiting periods, reprinting, paper and binding waste, transport-related emissions, and the need for physical storage. It also shows that the same report is duplicated across several locations, making retrieval, preservation, and administrative handling less efficient.

For a clearer view and to see the simulation click here:

https://drive.google.com/file/d/1USfY7Y5E0nqdssusB6i2B-IXnDzUK8Kf/view?usp=sharing

TO-BE MODEL



The TO-BE model presents the proposed paperless Capstone C submission process through a Digital Capstone Repository. The student logs in using an EWU Gmail account, enters the required project information, and uploads one final thesis PDF. The system automatically checks the file format and metadata. If any error is detected, the student receives an error message, corrects the report, and resubmits it digitally.

After successful validation, the report is sent to the supervisor for online review. The supervisor can approve the report or reject it with comments. When rejected, the student receives the feedback, makes the required corrections, and submits the revised report through the same system.

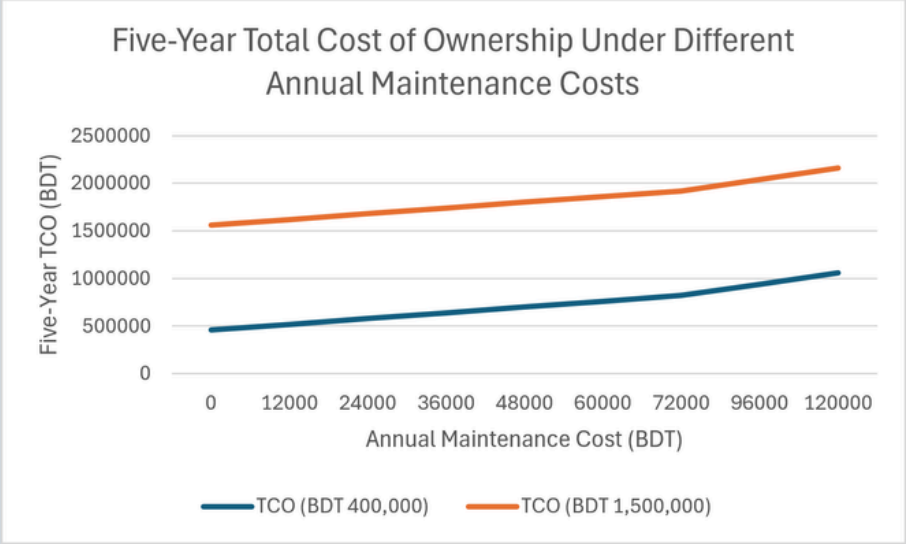
Once approved, the system locks the verified PDF, assigns a unique report ID, records the submission date and time, generates a file hash, and stores the report securely. The student then receives an acceptance notification, and the approved report becomes accessible to authorized EWU Gmail users.

This redesigned process removes compulsory printing, binding, travel, and physical storage. It reduces paper waste, waiting time, printing errors, student costs, and environmental impact while improving report verification, accessibility, preservation, and administrative efficiency.

For a clearer view and to see the simulation click here:
https://drive.google.com/file/d/1D3JrzqYvRl7DZJjSh57OmN1ghlPQwqwx/view?usp=sharing

ESTIMATED COST OF IMPLEMENTING THE TO-BE MODEL

The proposed Digital Capstone Repository would require an estimated development cost of approximately BDT 400,000 to BDT 1,500,000. This is considered a reasonable production-level estimate for a medium-complexity system containing role-based permissions, automated workflows, reporting features, and external integrations. A student prototype could be developed at a lower cost, but it would not provide the same level of security, reliability, scalability, testing, and technical support required for an official university system.



A Bangladesh-hosted 20 GB advanced hosting package would cost approximately BDT 12,000 per year. The package includes free SSL, SSH access, and up to seven days of daily backups. EWU could use a university subdomain, such as capstone.ewubd.edu, which would remove the need to purchase a separate domain.

The repository's annual running cost would consist of the BDT 12,000 hosting charge plus annual maintenance expenses. Maintenance costs are not fixed because they depend on university IT support, security updates, database administration, bug fixes, backup monitoring, user-account support, and future system modifications. Therefore, the annual running cost can be expressed as BDT 12,000 plus the annual maintenance cost.

Over five years, the total cost of ownership would depend on both the initial development cost and the selected maintenance level. With an initial development cost of BDT 400,000, the estimated five-year cost ranges from BDT 460,000 when no separate maintenance cost is included to BDT 1,060,000 when annual maintenance is BDT 120,000. With an initial development cost of BDT 1,500,000, the estimated five-year cost ranges from BDT 1,560,000 to BDT 2,160,000 under the same maintenance scenarios. These figures show that the repository requires a significant initial investment, but it provides a long-term institutional system for secure submission, approval, storage, retrieval, and preservation of Capstone reports.

ENVIRONMENTAL FOOTPRINT

ASSUMPTIONS

The environmental footprint calculation assumes that the Digital Capstone Repository operates continuously for one four-month semester, equivalent to approximately 2,920 hours. The repository is assumed to use one continuously running virtual CPU and 20 GB, or 0.02 TB, of SSD storage. A Power Usage Effectiveness value of 1.25 is used to account for cooling, lighting, power conversion, and other data-centre overhead. This value was selected as a reasonable conservative assumption

between the PUE values of approximately 1.10 for Google data centres and 1.39 for colocated data centres reported in Etsy's Cloud Jewels methodology. The calculation also assumes that 50 Capstone reports are submitted during each semester.

ESTIMATED ELECTRICITY FOOTPRINT

Etsy's Cloud Jewels methodology estimates cloud-computing electricity use at 2.10 Wh per vCPU-hour for computing and 1.52 Wh per TB-hour for SSD storage. Based on 2,920 operating hours, one virtual CPU would consume approximately 6.132 kWh of electricity, while 20 GB of SSD storage would consume approximately 0.089 kWh. The total IT electricity consumption would therefore be approximately 6.221 kWh per semester. After applying the PUE value of 1.25 to include data-centre overhead, the repository's total electricity footprint is estimated at approximately 7.78 kWh per four-month semester.

ESTIMATED CARBON FOOTPRINT

The repository's carbon footprint is calculated using the Bangladesh grid-emission factor of 0.62 kg CO_2e per kWh. Applying this factor to the estimated total electricity consumption of 7.776 kWh results in approximately 4.82 kg CO_2e per semester. This estimate represents the operational carbon emissions associated with running the repository and does not include the embodied emissions from manufacturing servers, network equipment, or users' devices.

ESTIMATED WATER FOOTPRINT

The repository has an indirect water footprint because data centres may consume water for cooling, while electricity generation also requires water. A direct data-centre cooling-water factor of 0.56 litres per kWh of IT electricity is used, producing an estimated cooling-water footprint of approximately 3.48 litres per semester. An indirect electricity-generation water factor of 1.04 litres per kWh is applied to the total data-centre electricity consumption, resulting in approximately 8.09 litres per semester. The combined operational water footprint is therefore estimated at approximately 11.57 litres per four-month semester. This should be described as a conservative illustrative estimate because actual water consumption depends on the data-centre location, cooling system, electricity source, and hosting provider.

ESTIMATED ELECTRICITY COST

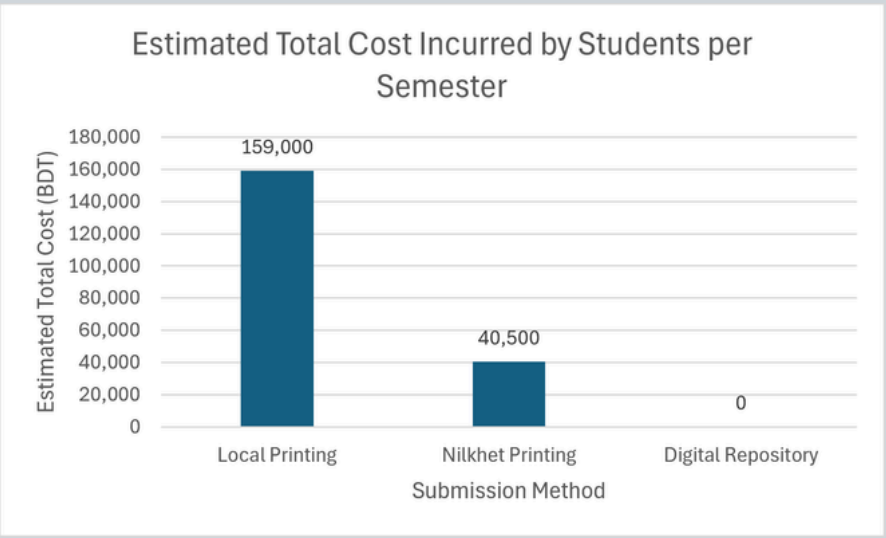
Using the Bangladesh commercial and office electricity tariff of BDT 15.36 per kWh, the estimated electricity component of operating the repository would be approximately BDT 119.44 per semester. For an on-campus server using the educational-institution tariff of BDT 9.05 per kWh, the corresponding cost would be approximately BDT 70.37 per semester. For a commercially hosted repository, electricity would normally be included within the annual hosting charge rather than appearing as a separate electricity bill.

ESTIMATED ENVIRONMENTAL FOOTPRINT PER REPORT

Assuming that 50 Capstone reports are submitted per semester, the repository would consume approximately 0.156 kWh of electricity per report. The estimated carbon footprint would be approximately 0.096 kg CO_2e per report, while the indirect water footprint would be approximately 0.231 litres per report. These per-report figures show that the operational environmental impact of storing and processing each digital report is relatively small.

AS-IS VS TO-BE MODEL

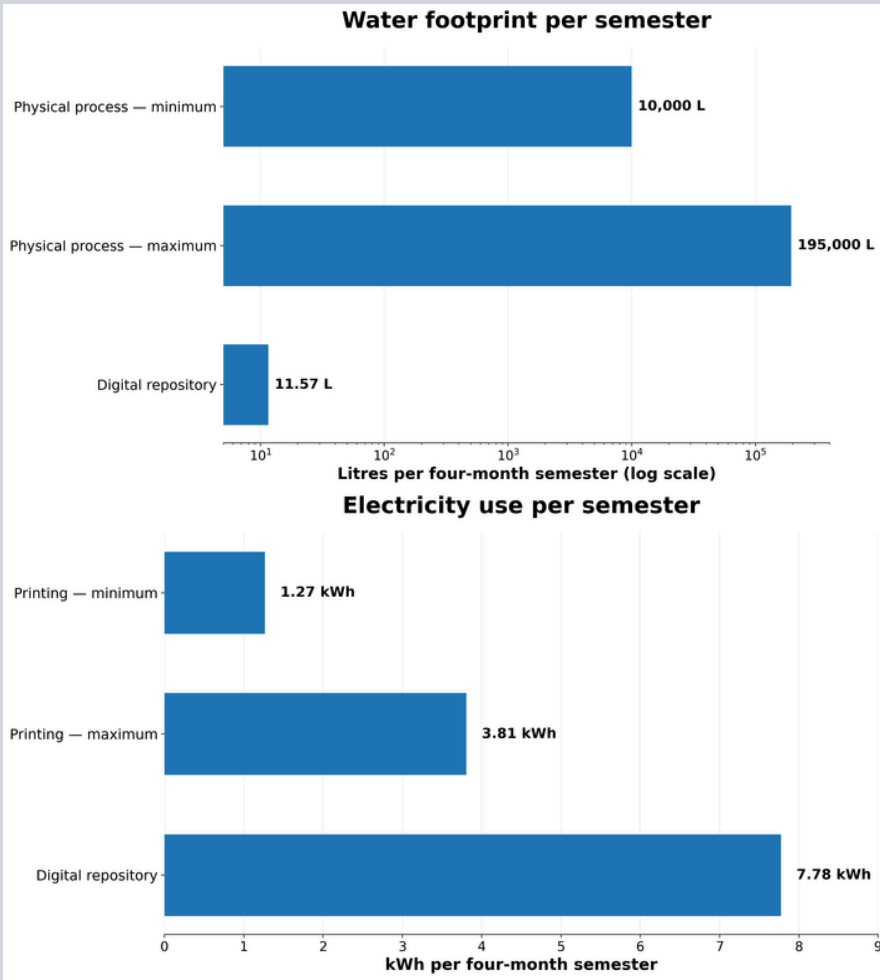
Comparison area	Physical submission process	Digital repository
Submission format	Two or three printed and bound copies	One verified digital PDF
Student printing cost	BDT 13,500–159,000 per semester	Almost zero compulsory printing cost
University investment	No new system-development cost	BDT 400,000–1,500,000 initial development
Annual hosting	Not applicable	Approximately BDT 12,000 plus maintenance
Paper use	5,000–15,000 sheets	No compulsory paper
Electricity use	1.27–3.81 kWh per semester	Approximately 7.78 kWh per semester
Carbon footprint	Approximately 160.15–228.85 kg CO_2e under the 16-km travel scenario	Approximately 4.82 kg CO_2e
Water footprint	Approximately 10,000–195,000 litres	Approximately 11.57 litres
Travel	Travel to a printing shop may be required	Online submission
Errors	Reprinting may waste paper, time, and electricity	Digital correction and resubmission
Storage	Physical shelves and multiple copies required	Central searchable storage
Accessibility	Limited by the location of physical copies	Available to authorized EWU users
Overall assessment	Lower system investment but high recurring environmental and student burden	Higher initial investment but much lower overall environmental impact and better long-term efficiency

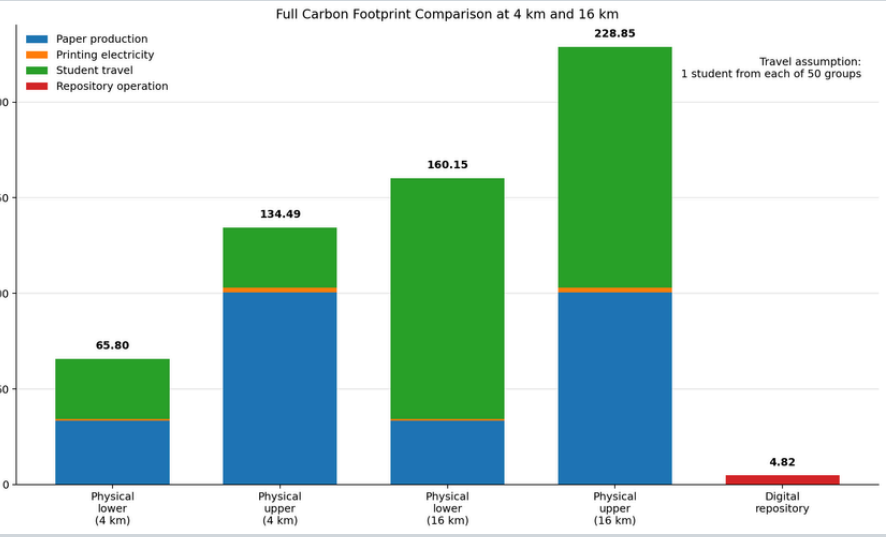


The graph shows that the current physical submission process creates a significant recurring cost for students. Local printing may cost up to BDT 159,000 per semester, while Nilkhet printing may cost around BDT 40,500. These estimates include printing and binding only and exclude transportation costs, travel time, waiting time, and possible reprinting expenses. With the proposed digital repository, students would have almost no compulsory printing or binding cost, although the university would bear the development, hosting, and maintenance costs.

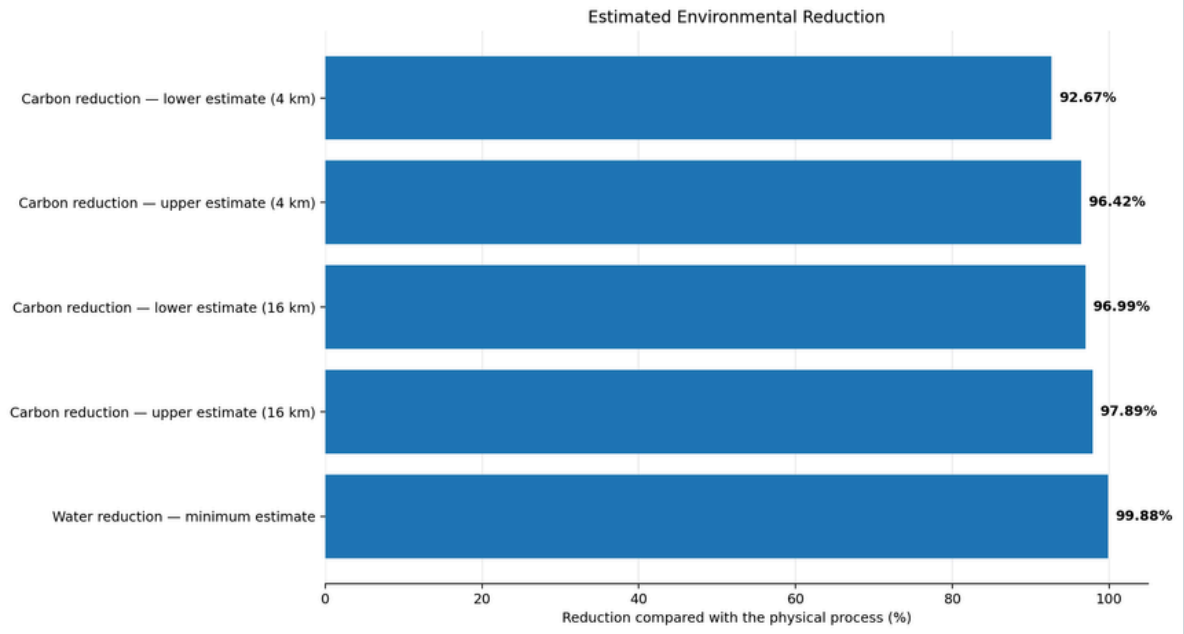
This graph compares the water footprint of the current physical process with the proposed digital repository. Producing the required paper may use approximately 10,000 to 195,000 litres of supply-chain water per semester, depending on the paper source and production method. In contrast, the repository's estimated indirect water footprint is only about 11.57 litres, mainly from data-centre cooling and electricity generation. Therefore, the digital process greatly reduces the overall water footprint, even though it is not completely impact-free.

This graph compares the electricity use of the two processes. Active printing consumes about 1.27 to 3.81 kWh per semester, while the digital repository is estimated to use about 7.78 kWh because it operates continuously and includes data-centre overhead. Therefore, the repository does not reduce electricity use by itself. However, it is greener overall because it avoids the much larger impacts of paper production, printing, binding, reprinting, and student travel.





This graph compares the total carbon footprint of the physical and digital processes. At a 4-kilometre return distance, the physical process produces about 65.80 to 134.49 kg CO_2e, while at 16 kilometres it increases to about 160.15 to 228.85 kg CO_2e because of additional travel emissions. In comparison, the digital repository produces only about 4.82 kg CO_2e per semester. Therefore, the digital process has a much lower overall carbon footprint, especially as student travel distance increases.



This graph shows the estimated environmental reduction achieved by the TO-BE model. At a 4-kilometre return distance, carbon emissions decrease by approximately 92.67% to 96.42%, while at 16 kilometres the reduction increases to about 96.99% to 97.89%. The estimated water-footprint reduction is around 99.88%. This shows that the digital repository provides greater environmental benefits as the avoided travel distance increases.

MOTIVATION FOR UNIVERSITIES TO ADOPT THE TO-BE MODEL

The university should consider introducing a Digital Capstone Repository because it would solve several practical problems at once.

1. Instead of requiring students to print and bind three identical copies, the university could keep one approved digital version that is easy to verify, search, store, and access.
2. Supervisors and the chairperson could approve the report online, while the department and library could retrieve the same verified copy whenever needed.
3. This would reduce paper use, student travel, storage pressure, and the time staff spend handling physical books.
4. It would also help during accreditation and departmental reviews because reports could be found within seconds rather than searched for manually.
5. A well-managed repository would also improve the university's brand value by showing that it supports research, uses technology responsibly, and takes sustainability seriously.

DETAILED CALCULATIONS

FOR THE AS-IS MODEL

Here the assumption is every semester about 200 students take capstone C and assuming every group has 4 members, there are 50 groups and so 50 reports.

Carbon emission:
Weight of one A4 sheet:
W_sheet (0.210 * 0.297 sized paper) = 0.210 × 0.297 × 80 = 4.9896 g
Weight of 5,000 sheets:
W_5000 = (5000 × 4.9896) / 1000 = 24.948 kg ≈ 25 kg
Weight of 15,000 sheets:
W_15000 = (15000 × 4.9896) / 1000 = 74.844 kg ≈ 75 kg
Paper-emission factor: (Using the 2026 UK Government primary-paper material factor)
EF_paper = 1343.59114 / 1000 = 1.34359114 kg CO_2e/kg
Carbon emissions from 5,000 sheets:
E_5000 = 24.948 × 1.34359114 = 33.52 kg CO_2e ≈ 34 kg CO_2e
Carbon emissions from 15,000 sheets:
E_15000 = 74.844 × 1.34359114 = 100.57 kg CO_2e ≈ 101 kg CO_2e
Assuming standard 80-gsm A4 paper, 5,000–15,000 sheets equal approximately 25-75 kg of paper and create an estimated paper-production footprint of about 34 - 101 kg CO_2e.

Printer power:
The HP Color LaserJet Enterprise 5800zf uses 655 W while printing and prints up to 43 A4 pages per minute.
P = 655 / 1000 = 0.655 kW
Printer speed:
S = 43 pages/minute
The current Bangladesh commercial and office electricity tariff is:
C_e = ৳15.36/kWh
For 5,000 pages:T_5000 = 5000 / (43 × 60) = 1.938 hours
E_5000 = 0.655 × 1.938 = 1.269 kWh
C_5000 = 1.269 × 15.36 = ৳19.50
CO_2_5000 = 1.269 × 0.648 = 0.823 kg CO_2
For 15,000 pages: T_15000 = 15000 / (43 × 60) = 5.814 hours
E_15000 = 0.655 × 5.814 = 3.808 kWh
C_15000 = 3.808 × 15.36 = ৳58.49
CO_2_15000 = 3.808 × 0.648 = 2.468 kg CO_2
Final range:
Electricity consumption = 1.269–3.808 kWh per semester
Electricity cost = ৳19.50–৳58.49 per semester
Printing-electricity emissions = 0.823–2.468 kg CO_2 per semester

Water Footprint

The Water Footprint Network estimates that producing printing and writing paper requires approximately 2–13 litres of supply-chain water per A4 sheet, depending on the paper's source, type, recycled content, and production process. This includes water used in forestry and paper manufacturing, not only water used at the printing shop.

Minimum water footprint for 5,000 sheets:

$W_{min} = 5{,}000 \times 2 = 10{,}000$ L

Maximum water footprint for 5,000 sheets:

$W_{max} = 5{,}000 \times 13 = 65{,}000$ L

Water-footprint range for 5,000 sheets:

$W_{5{,}000} = 10{,}000–65{,}000$ L

Minimum water footprint for 15,000 sheets:

$W_{min} = 15{,}000 \times 2 = 30{,}000$ L

Maximum water footprint for 15,000 sheets:

$W_{max} = 15{,}000 \times 13 = 195{,}000$ L

Water-footprint range for 15,000 sheets:

$W_{15{,}000} = 30{,}000–195{,}000$ L

Overall range:

$W_{overall} = 10{,}000–195{,}000$ L

The paper's supply-chain water footprint could range from approximately 10,000 to 195,000 litres, depending on the source and production process.

Travelling

The UK Government's 2026 conversion factors report 0.12552 kg CO_2e per passenger-kilometre for a non-London local bus.

Including the fuel supply or well-to-tank factor of 0.03174 kg CO_2e per passenger-kilometre, the combined planning factor is:

$EF_{bus} = 0.12552 + 0.03174 = 0.15726$ kg CO_2e/passenger-km

Number of Capstone groups:

$N = 50$ groups

Return journey per student (just assuming):

$D_{return} = 16$ km

For one student from each group:

$D_{total,1} = 50 \times 1 \times 16 = 800$ passenger-km

$E_{travel,1} = 800 \times 0.15726 = 125.808$ kg $CO_2e \approx 126$ kg CO_2e

Using the UK Government's 2026 conversion factors report 0.12552 kg CO_2e per passenger-kilometre for a non-London local bus. A current, authoritative Bangladesh source that directly provides a complete local-bus factor in kg CO_2e per passenger-kilometre was not found. Bangladesh documents provide national transport emissions and some Dhaka-specific inputs, but not a ready-to-use passenger-kilometre factor.

FOR THE TO-BE MODEL

The calculation uses the supplied coefficients and assumes:

- One four-month semester = approximately 2,920 operating hours
- One continuously running vCPU
- SSD storage capacity = 20 GB = 0.02 TB
- PUE = 1.25

Etsy reported a PUE of approximately 1.10 for Google's data centres and 1.39 for its former colocated data centres. Therefore, PUE = 1.25 was selected as a reasonable conservative assumption between these values.

- 50 Capstone reports per semester

Computing Electricity

Etsy's Cloud Jewels methodology estimates cloud-computing electricity consumption using:

- 2.10 Wh per vCPU-hour for server computing
- 1.52 Wh per TB-hour for SSD storage

Compute electricity:

$E_{compute}$ = 2.10 Wh × 2,920 hours

$E_{compute}$ = 6,132 Wh

$E_{compute}$ = 6.132 kWh

2. Storage Electricity

$E_{storage}$ = 1.52 Wh × 0.02 TB × 2,920 hours

$E_{storage}$ = 88.768 Wh

$E_{storage}$ = 0.089 kWh

Total IT Electricity

$E_{IT} = E_{compute} + E_{storage}$

E_{IT} = 6.132 + 0.089

E_{IT} = 6.221 kWh per semester

4. Electricity Including Data-Centre Overhead

PUE includes cooling, lighting, power conversion and other supporting infrastructure.

$E_{total} = E_{IT} \times PUE$

E_{total} = 6.221 × 1.25

E_{total} = 7.776 kWh per semester

Therefore:

Electricity footprint = approximately 7.78 kWh per four-month semester

Carbon Footprint
Bangladesh grid-emission factor:
$EF_{grid} = 0.62$ kg CO_2e/kWh
Carbon footprint:
$CO_2e = E_{total} \times EF_{grid}$
$CO_2e = 7.776 \times 0.62$
$CO_2e = 4.821$ kg CO_2e per semester
Therefore:
Carbon footprint = approximately 4.82 kg CO_2e per semester

Direct Data-Centre Cooling Water
Direct water factor:
$WUE_{direct} = 0.56$ litres/kWh of IT electricity
$W_{direct} = E_{IT} \times WUE_{direct}$
$W_{direct} = 6.221 \times 0.56$
$W_{direct} = 3.484$ litres per semester
Indirect Electricity-Generation Water
Indirect water factor:
$WUE_{indirect} = 1.04$ litres/kWh
$W_{indirect} = E_{total} \times WUE_{indirect}$
$W_{indirect} = 7.776 \times 1.04$
$W_{indirect} = 8.087$ litres per semester
Total Water Footprint
$W_{total} = W_{direct} + W_{indirect}$
$W_{total} = 3.484 + 8.087$
$W_{total} = 11.571$ litres per semester
Therefore:
Water footprint = approximately 11.57 litres per semester

Footprint Per Capstone Report
Assuming 50 reports per semester:
Electricity per report:
$7.776 \div 50 = 0.156$ kWh
Carbon per report:
$4.821 \div 50 = 0.096$ kg CO_2e
Water per report:
$11.571 \div 50 = 0.231$ litres

CARBON FOOTPRINT CALCULATION

Carbon footprint of the AS-IS model (using distance = 16km)
The physical-process carbon footprint includes:
Paper-production emissions + printing-electricity emissions + travel emissions
For 5,000 pages:
Paper emissions = 33.52 kg CO_2e
Printing-electricity emissions = 1.269 × 0.62 = 0.79 kg CO_2e
Travel emissions for a 16-km return journey = 125.81 kg CO_2e
Total lower estimate:
33.52 + 0.79 + 125.81 = 160.11 kg CO_2e per semester
For 15,000 pages:
Paper emissions = 100.56 kg CO_2e
Printing-electricity emissions = 3.808 × 0.62 = 2.36 kg CO_2e
Travel emissions = 125.81 kg CO_2e
Total upper estimate:
100.56 + 2.36 + 125.81 = 228.73 kg CO_2e per semester
Therefore:
AS-IS carbon footprint = approximately 160.11–228.73 kg CO_2e per semester

Carbon footprint of the TO-BE model
Repository electricity consumption = 7.776 kWh
Digital-repository carbon emissions:
7.776 × 0.62 = 4.82 kg CO_2e per semester

Carbon-emission reduction
Lower scenario:
160.11 − 4.82 = 155.29 kg CO_2e avoided
Percentage reduction:
(155.29 ÷ 160.11) × 100 = 96.99%
Upper scenario:
228.73 − 4.82 = 223.91 kg CO_2e avoided
Percentage reduction:
(223.91 ÷ 228.73) × 100 = 97.89%
Therefore:
Estimated carbon avoided = 155.29–223.91 kg CO_2e per semester
Estimated carbon-emission reduction = approximately 97–98%

Scenario	Paper emissions	Printing emissions	Travel emissions	Total AS-IS carbon	TO-BE carbon	Carbon reduction
4 km, lower estimate	33.52	0.79	31.45	**65.76 kg CO_2e**	4.82 kg CO_2e	**92.67%**
4 km, upper estimate	100.56	2.36	31.45	**134.37 kg CO_2e**	4.82 kg CO_2e	**96.41%**
16 km, lower estimate	33.52	0.79	125.81	**160.11 kg CO_2e**	4.82 kg CO_2e	**96.99%**
16 km, upper estimate	100.56	2.36	125.81	**228.73 kg CO_2e**	4.82 kg CO_2e	**97.89%**

FINAL INTERPRETATION

The digital repository does not reduce electricity consumption, because its estimated electricity use is higher than active printer electricity alone. However, it is greener overall because it avoids the much larger impacts associated with paper production, supply-chain water use, binding, reprinting, and printing-related travel.

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